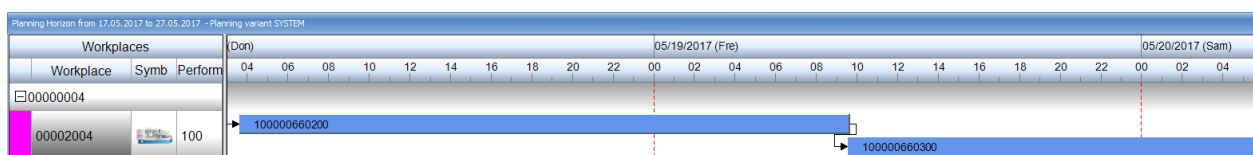


1 Configuration of overlapping operations

Overview

You commonly define dependencies between operations (OPs) as finish-to-start relationships. When scheduling operations, the finish-to-start relationship has the effect that the succeeding OP cannot be started (change from status *prepared* to status *running*) until the preceding OP is finished 100 % (status *finished*). In the HYDRA shop floor scheduling (HLS), operations that are dependent on each other must never overlap; this is shown in the following screenshot:



You can change existing dependencies using the "overlapping" function. With the overlapping function, you can schedule the succeeding OP even if the preceding OP has not yet been finished 100 % and is still in status *running*. An overlapping might be useful if the succeeding OP can start with a part quantity that has been produced by the preceding OP.

The necessary configurations for an overlapping of two operations are described in the following.

Configurations in the *Processing code*

Menu	Master data → Order → Processing codes
Transaction code	Pc
Function authorization	Mdpc

Each operation is based on a processing code. You can find the processing code of an existing operation in the application *Order information*, tab *Operations*, sub-tab *Processing*.

If you want to enable overlapping, configure the processing code of the respective operations as follows.

1. Go to the application *Processing codes*.
2. Select the processing code for which you want to configure overlapping and click the button *Edit*.
3. Go to the tab *Planning* and select the entry *S Target overlapping* of the drop-down menu in the field *Overlapping*.

Further information on processing codes and the configuration possibilities can be found [here](#).

Configurations in the *Operation*

Menu	Order Management → Order Management → Edit Operations
Transaction code	edop
Function authorization	edop

By configuring the processing code you have enabled overlapping. In the application *Operation*, you must configure if and when the overlapping of two operations should take place.

1. Go to the application *Edit operations*.
2. Select the preceding OP and edit it.

Two settings are relevant for the overlapping:

- Send-ahead quantity
- Synchronization time

Change to the tab *Quantities*. In the field *Send-ahead quantity*, enter the quantity needed to start with the succeeding OP.

This configuration has the effect that e.g. in the *Graphic planning* the succeeding OP can be scheduled free of conflicts for the (calculated) point in time when the preceding OP will have produced the send-ahead quantity. The calculation of the point in time when the preceding OP will have produced the send-ahead quantity is based on the value stored in the field *Target processing time* in the tab *Durations*.

The calculation of the time when the send-ahead quantity is reached always depends on the value in the field *Target processing time*. This also applies if you want to identify the operation time/remaining run time in the *Graphic planning* using the target cycle.



The formula stored in the field *Processing time formula* is used to calculate the value that is displayed in the field *Target processing time*.

If you want to avoid that a succeeding OP starts too early despite a sufficient send-ahead quantity of the preceding OP, you must configure a maximum synchronization time. The maximum synchronization time is the time the preceding OP must have been processed at least before the succeeding OP can be scheduled free of conflicts. A possible configuration of the preceding OP would be as follows:

- Target quantity: 10,000 pieces
- Target processing time: 30 hours
- Send-ahead quantity: 1,000 pieces

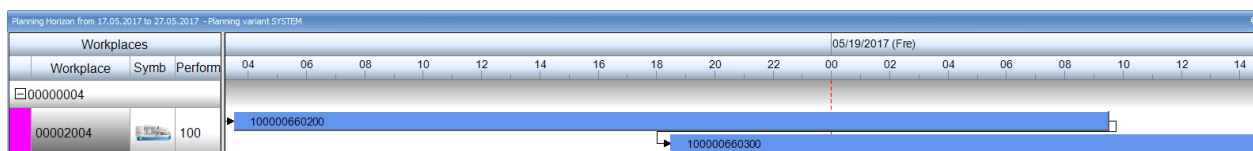
➔ The succeeding OP could start after 3 hours.

- Maximum synchronization time: 10 hours
 - ➔ The succeeding OP is only scheduled after the preceding OP has been processed for 10 hours.

Effects of the configurations

Menu	Production control → Production preparation → Graphic planning
Transaction code	grap
Function authorization	grap

You have enabled an overlapping of two operations by way of the above mentioned configurations. For example, the *Graphic planning* is affected by the configurations made. Here, you can now plan overlapping operations free of conflicts by manual assignment or via automatic assignment as shown in the screenshot below.



In addition, a start-to-start relationship is established for the operations that are affected by the overlapping. In case of a start-to-start relationship, the start of the succeeding OP depends on the start of the preceding OP.

Edit order network						
Drag a column here to group by that column						
Predecessor		Successor		Options		
Predecessor	Preceding OP	Successor	Succeeding OP	Active	Relationship	Origin
10000066	0100	10000066	0200	☒	ES	S
> 10000066	0200	10000066	0300	☒	SS	S
10000066	0200	10000066	0300	☒	ES	S
10000066	0300	10000066	0400	☒	ES	S
10000066	0400	10000066	0500	☒	ES	S
10000066	0500	10000066	0600	☒	ES	S
10000066	0600	10000066	0700	☒	ES	S